

PXIe based vector signal analysers (VSAs), namely, M9391A or M9393A, and Keysight's 89600 VSA software, which provide digital demodulation and spectrum-analysis capabilities, constitute the satellite-monitoring reference solution. Modular architecture based hardware provides engineers the scalability to meet a variety of applications and upgrade as needs change in the future.

Conventionally, multiple analysers are required to monitor several bands in the sequential scanning satellite signal-monitoring configuration. But with M9393 PXIe VSA, diverse bands can be monitored at once. It also allows for quick measurements with multi-band scanning and display refresh rates exceeding 280GHz per second. Users can customise the signal monitoring solution by utilising the measurement setup files as well as test code logging and 89600 VSA software automation layer examples, thereby allowing consolidation in the test environment. With 89600 VSA software, engineers can evaluate modulation, time and complex signal spectrum characteristics. They can also implement spectrum scanning and zoom in on signals using digital vector analysis as wide as 160MHz.

Analysing, monitoring system combo for mission-critical satellite bandwidth

A network-carrier monitoring system is used by satellite operators and government agencies for standard as well as tactical applications for limit-less monitoring capabilities. Such systems help protect mission-critical satellite bandwidth and assist in monitoring spectrum and data from extensive remote sites.

With Anritsu making their spectrum/signals analyser MS2830A compatible with SAT Corp.'s Monics (a satellite carrier monitoring system software), interference detection and automated monitoring is made easier to achieve.

In addition to time-domain measurements and signal-modulation

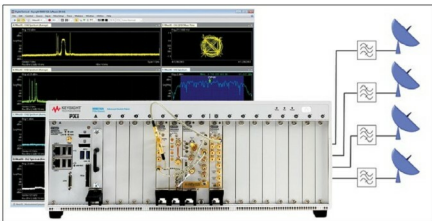


Fig. 2: Scanning multiple bands (satellite dishes) with a single analyser (Image courtesy: www.keysight.com)

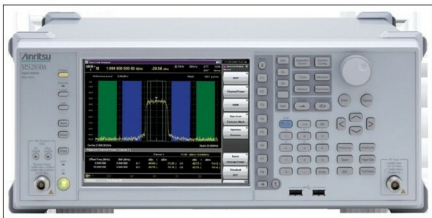


Fig. 3: Anritsu's MS2830A spectrum analyser/signal analyser is compatible with Monics satellite carrier monitoring system from SAT Corp. (SAT) (Image courtesy: www.anritsu.com)

analysis, engineers can also perform spectrum radio frequency (RF) measurements such as centre frequency, carrier-to-noise ratio (C/N), bandwidth and equivalent isotropically-radiated power (EIRP), to name a few. With the installation of a driver, MS2830A spectrum/signal analyser can be integrated into the Monics enterprise-networked carrier-monitoring system to provide real-time data for uplink and downlink of satellite communication. Based on stored reference traces and familiar carrier plans, interference too can be detected using this analyser.

Monics, on the other hand, ex-

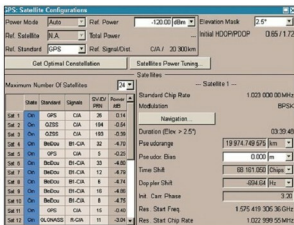


Fig. 4: The R&S SMBV100A generates hybrid GPS, Glonass, Galileo, BeiDou and QZSS satellite constellations with minimum position dilution of precision (Image courtesy: www.rhde-schwarz.co.in)

tends interference detection and signal characterisation. It provides simultaneous manual and automatic monitoring with an easy and efficient user interface. The automatic monitoring compares the expected and